

5 Interesting Reference Designs For DISPLAY AND GUI APPLICATIONS

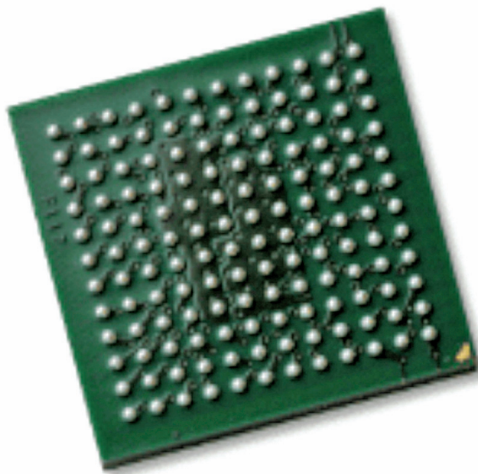
Building small, high-quality displays from scratch can be complex and time-consuming. These reference designs manage video, colour, and optics, reducing design time and effort.

NIDHI AGARWAL

When we develop displays for compact devices such as wearables, touchscreens, or automotive screens, integration often adds complexity and delays. These designs provide ready-made hardware and software, allowing us to bypass ground-up development. As a result, we can build reliable displays faster, cut engineering effort, and minimise errors.

Ultra-compact wearable display

This reference design provides a compact, ready-to-use display system, eliminating the need to build a micro-display setup from scratch. It includes a 720p display engine



and a smart controller that manages video input, scaling, frame rate, colour, lighting timing, image correction, and optics within a small footprint. This cuts design risk and engineering time, while ensuring reliable, low-power operation. Built-in firmware, a microcontroller, a frame buffer, a light module, and integration guides enable us to quickly develop clear, high-quality displays for AR glasses, viewfinders, medical scopes, inspection tools, compact projectors, and

other small imaging devices where size, power efficiency, and image clarity are critical.

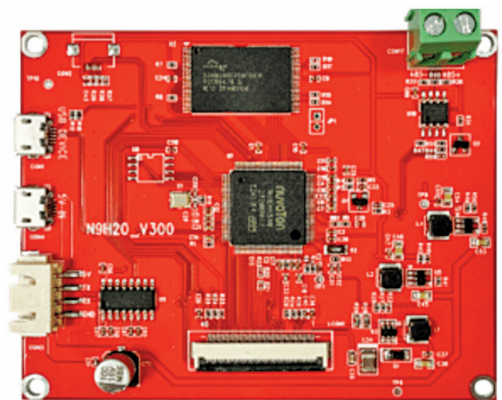
Applications. Suitable where size, power efficiency, and visual clarity are paramount

Link. <https://www.electronicsforu.com/electronics-projects/ultra-compact-wearable-display-reference-design>

OEM Brand. Omnivision Technologies

GUI for embedded display applications

The GUI reference design portfolio provides ready-to-use hardware designs for embedded products with graphical user interfaces (GUIs). It provides a



complete, pre-validated starting point, allowing us to avoid building display-subsystem hardware from scratch. The main problem it solves is making a TFT-LCD display work with a controller while keeping all display, touch, and peripheral interfaces working together. We can quickly prototype systems, connect standard interfaces such as USB, UART, CAN, or Ethernet to sensors or networks, and evaluate performance early in development. This reduces risk and shortens development cycles, allowing us to focus